

42.510 Statistical Learning for Data Science

Linear regression

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Week 2, Lecture 2

Bordeaux wines

- Points to ponder
 - ▶ Excellent weather conditions conducive for growing vines.
 - ▶ Difference in price and quality between years.
 - ▶ Aged wine is supposed to taste good.
Younger wines not so pleasant to taste.
- Hence the importance of wine experts.
 - ▶ Robert Parker

Vintage chart 1970-2023

Robert Parker Wine Advocate
Vintage Chart 1970 - 2023

AUSTRIA

AUSTRIA	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003	2002	2001	2000	1999	
AUSTRIA: NIEDERÖSTERREICH / WIEN / BURGENLAND WHITE			NT	92- 94E	90- 97T	93- 93T	92R	90R	95T	78E	91R	89T	89E	88I	89I	88R	90R	91I	87I	88I	89I	89T	88I	85C	95R	
AUSTRIA: STYRIA (STEEIERMARK)			NT	NT	90- 93E	92- 94R	91- 93R	96E	90- 92R	95R	87- 88E	89- 94R	90- 93R	NT	NT	NT	NT	NT	NT	NT	NT	NT	NT	NT	NT	
AUSTRIA: NIEDERÖSTERREICH / WIEN / BURGENLAND RED			NT	NT	NT	94- 96T	89- 91E	93- 95T	90- 92R	91- 94T	79- 85I	90- 92R	90- 94R	93- 95R	88- 94I	93- 94R	75- 80R	90R	93- 95R	NT	88R	88- 92R	89- 92R	86- 90I	89- 92R	94R

FRANCE

ALSACE		NT	NT	89-92T	94-96T	89-91E	93T	90E	94T	89I	86T	87R	86E	89I	82I	89R	91R	79I	87R	86E	82R	86R	91R	90R	87R
BORDEAUX: ST JULIEN/PAUILLAC/ST ESTEPHE	94	95T	89I	96T	96T	94T	91I	97T	93E	95T	81C	92E	88E	98T	99E	91E	86E	87E	95T	88T	95T	88T	88R	96T	88R
BORDEAUX: MARGAUX	93	95T	88I	95T	96T	94T	90I	97T	94E	93T	80C	89E	87E	95T	97E	90E	86E	88E	98T	87T	88I	88T	89E	94T	89R
BORDEAUX: GRAVES/PESSAC LEOGNAN	94		89I	95T	96T	94T	92I	97T	95E	93T	81C	91E	86E	99T	98E	91E	87E	87E	96T	88T	88I	87T	88R	97T	88R
BORDEAUX: POMEROL	94	95T	87I	95T	96T	94T	90I	97T	95E	94T	84R	94E	88R	95T	98E	96E	86I	90T	95T	88E	84E	85E	90E	95T	88R
BORDEAUX: ST EMILION	95	95T	88I	95T	95T	94T	89I	96T	95E	92T	82R	93E	87R	94T	93E	92E	86I	88E	99T	88E	90I	87E	90E	96T	88R
BORDEAUX: BARSAC/AUTERNES	NT				92E	90E	91E	92E	95E	92E	92E	88E	93E	90R	97E	89R	94R	88R	96R	87R	89R	85R	93R	88E	88E
BURGUNDY: COTE DE NUITS (RED)	NT	95	92E	97T	98E	93T	93E	97T	98T	92E	92E	95T	91E	96E	95E	88I	91E	89I	98T	83C	89R	94R	91R	87R	92R
BURGUNDY: COTE DE BEAUNE (RED)	NT	95	91E	95T	96E	94T	92E	92E	96T	91E	89E	91E	90E	94E	95E	89I	90E	82I	96T	79C	87T	92R	88R	86R	93R
BURGUNDY (WHITE)	NT	96E	91T	97E	93T	91E	97E	87I	94E	97E	90E	92E	91E	94E	91E	94E	91I	90E	88E	91R	84R	92R	86C	88C	89C
BURGUNDY: BEAUJOLAIS	NT	91T	90I	93E	94E	93E	94E	95I	90I	93R	86R	90R	91R	93R	97R	86R	85R	89R	95R	81C	93R	86C	75C	91R	89R
CHAMPAGNE	NT	NT	86I	94T	95T	92E	87I	94T	93E	94E	95T	96T	87R	89R	92R	99T	80R	86R	88I	90T	88I	95T	88R	92R	92R
JURA	94T	91E	95E	92E	94E	93E	91I	93T	94T	93E	90I	93E	91R	94E	90R	91T	91I	90E	96R	92R	94R	93R	85R	NT	NT
LANGUEDOC		94	NT	92E	92T	91E	91T	92E	93T	87E	88E	88I	91R	94E	91R	87R	93R	90I	88R	88C	87I	NT	NT	NT	NT
ROUSSILLON		94	NT	92E	91T	90E	91T	92E	93T	88E	94E	89R	91R	94E	91R	87R	92R	90I	88R	88C	87I	NT	NT	NT	NT

2023 2022 2021 2020 2019 2018 2017 2016 2015 2014 2013 2012 2011 2010 2009 2008 2007 2006 2005 2004 2003 2002 2001 2000 1999

Vintage chart 1970-2023

RATINGS		
	96-100	Extraordinary
	90-95	Outstanding
	80-89	Above Average to Excellent
	70-79	Average
	60-69	Below Average
	<59	Appalling

MATURITY	
C	Caution, may be too old
E	Early maturing and accessible
NV	Vintage not declared
I	Irregular, even among the best wines
NT	Not yet sufficiently tested to rate
R	Ready to drink
T	Still tannic, youthful, or slow to mature

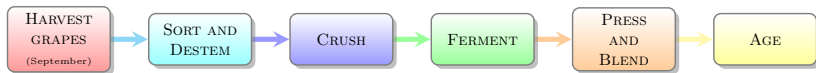
The Statistical Sommelier

- Orley Ashenfelter, Professor of Economics, Princeton University.
- Claims that he can predict quality without tasting.
- Tool used: Linear Regression.
- Liquid asset

Why predict?

- ... ???
- Investment: *en primeur* or *wine futures*
- tasting samples of wine within a year or two after it is made.
- The 1982 vintage wine of Chateau Latour that was sold at 250 pounds a case *en primeur* in 1983 was valued at 9000 British pounds in 2007.
- Video

The wine making process



Ashenfelter focused on the vintage as a predictor for the quality of wine by averaging auction prices across chateaus.

Observations

- Older the wine is, the greater is the value.
- There is still significant variation in average prices that remains unexplained.

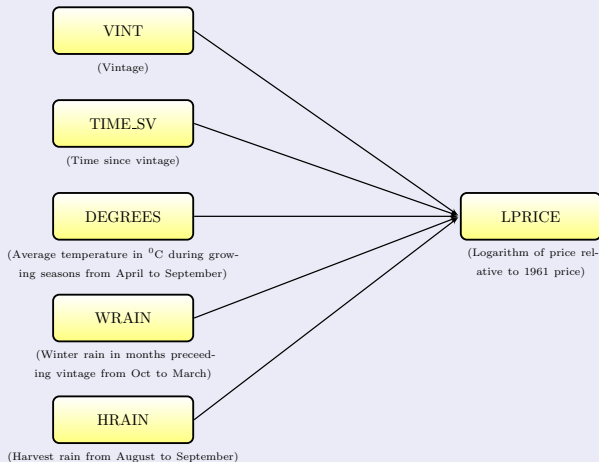
Possible predictors

- Vineyard (chateau - location where wine is made)
- Vintage (year - time when wine is made)

Wine making in Bordeaux

Datasets and variables

Years: 1952 - 1989



- Variables used from data in **Liquid asset**

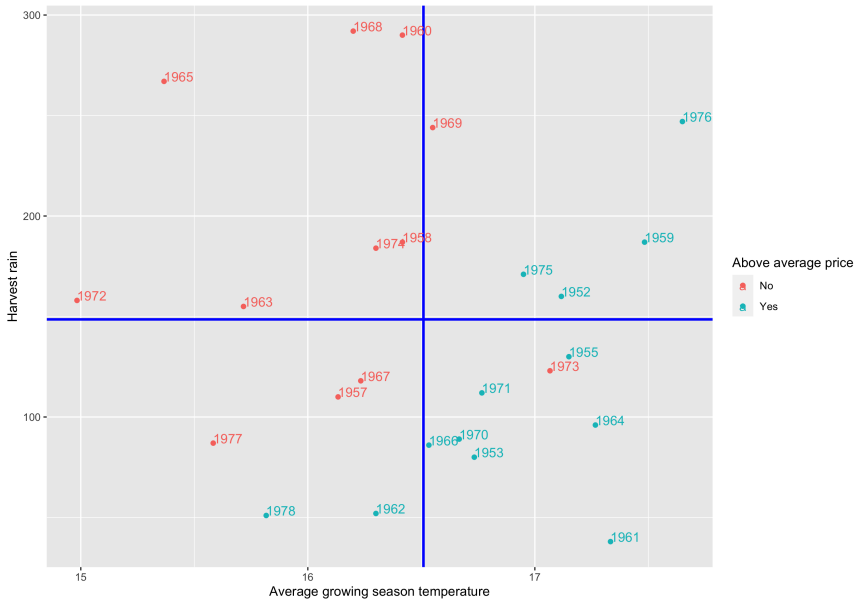
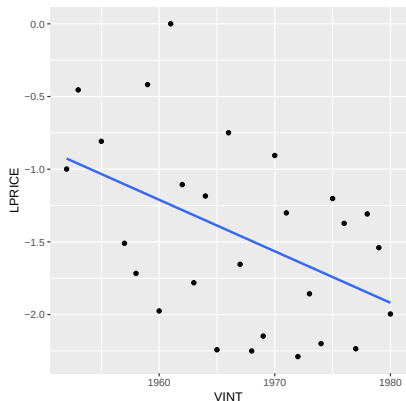


Figure 1: Harvest Rain vs Temperature

Linear regression



Problem setup

- n = Number of observations
- p = Number of predictor variables
- y = Dependent variable in $\mathbb{R}$
- $x_1, \dots, x_p$ = Independent variables (predictors)

- Want to predict y with a function $f(\mathbf{x})$.
- Let $f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$.
- Data: $(y_i, x_{i1}, \dots, x_{ip})$, $i = 1, \dots, n$.
- Method of least squares:

$$\min_{\beta_0, \dots, \beta_p} \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2.$$

- This is Empirical Risk minimization (ERM).
- Minimize loss L where

$$L(Y, f(\mathbf{X})) = \mathbb{E}(Y - f(\mathbf{X}))^2$$

with some assumptions on $\mathbb{P}(\mathbf{X}, Y)$.

- Structural assumption: $f(\mathbf{x}) = \mathbf{x}^\top \beta$.

Linear regression: Notation and set-up

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbb{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}.$$

- We want to minimize $Q(\boldsymbol{\beta})$ w.r.t $\boldsymbol{\beta}$ where

$$Q(\boldsymbol{\beta}) = (\mathbf{y} - \mathbb{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbb{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbb{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbb{X}^\top \mathbb{X} \boldsymbol{\beta}$$

- Taking derivatives and equating to zero we have obtain:

$$\hat{\boldsymbol{\beta}} = (\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top \mathbf{y}.$$

- *Fitted values:* $\hat{y}_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$.
- *Residuals:* $e_i = y_i - \hat{y}_i$.
- Note that $Q(\boldsymbol{\beta})$ is a quadratic in $\boldsymbol{\beta}$. This is a convex problem and hence a local minima is a global minima as well.

Linear regression

- No statistical assumptions until this. But we would like to compute standard error (variability in the estimates).
- Assume: $\mathbf{Y} = \mathbb{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$.
- Therefore $\text{Var}(\mathbf{Y}) = \text{Var}(\mathbb{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) = \text{Var}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}$.
- Since for any $k \times n$ matrix $\mathbf{A}$, we have $\text{Var}(\mathbf{A}\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$, we get

$$\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbb{X}^\top \mathbb{X})^{-1}$$

- Hence we have standard error: $\text{se}(\hat{\beta}_j) = \sigma \sqrt{((\mathbb{X}^\top \mathbb{X})^{-1})_{jj}}$.
- σ^2 is unknown but we can obtain an unbiased estimate of σ^2 from the data:

$$\hat{\sigma}^2 = \frac{1}{n - p - 1} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n - p - 1} \sum_{i=1}^n e_i^2 = \frac{Q(\hat{\boldsymbol{\beta}})}{n - p - 1}$$

where $Q(\hat{\boldsymbol{\beta}})$ is residual sum of squares (RSS) or sum of squared errors (SSE).

Linear regression: important questions

- Is X_i useful in predicting the response Y ?
- How well does the model fit the data?

Linear regression: Is the j -th variable significant?

- A natural question here: is the j^{th} variable useful in the model?
- We can conduct a one-sample t -test to ascertain this.
- We intend to test:

$$\mathbf{H}_0 : \beta_j = 0 \quad \text{vs.} \quad \mathbf{H}_1 : \beta_j \neq 0$$

- Since $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, we have obtained $\hat{\beta}_j \sim \mathcal{N}(0, \sigma^2 ((\mathbb{X}^\top \mathbb{X})^{-1})_{jj})$.
- Therefore under $\mathbf{H}_0$ we have the test statistic

$$t_j = \frac{\hat{\beta}_j}{\hat{\sigma} \sqrt{((\mathbb{X}^\top \mathbb{X})^{-1})_{jj}}} \sim t_{n-p-1},$$

a t -distribution with $(n - p - 1)$ degrees of freedom.

- We reject $\mathbf{H}_0$ for very high (absolute) values of the test statistic $|t_j|$ or very small p-values.

Linear regression: Quality of fit

- Note that for the regression model the following quantities are important:
- Sum of squared errors (SSE) or Residual sum of squares (RSS):

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Sum of squares due to regression (SSR):

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

- Total sum of squares (SST):

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

- We also have the following:

$$SST = SSR + SSE$$

Linear regression: Quality of fit

- Coefficient of Determination:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

- Check that $0 \leq R^2 \leq 1$.
- $R^2 = 0$: the linear regression model has no explanatory power.
- $R^2 = 1$: the linear regression model is a perfect predictor.
- If $Y = \beta_0 + \beta_1 X_1$ then $R^2 = (\text{corr}(X_1, Y))^2$.
- As the number of variables $p \uparrow$, we have $R^2 \uparrow$ as well.
- *Alternative measure*: Adjusted- R^2 :

$$\text{Adjusted-}R^2 = 1 - (1 - R^2) \frac{n - 1}{n - p - 1}$$

It penalizes for adding more variables.

Linear regression: other important questions

- Is the model significant?
- Are all variables important?
- How good is our prediction?
- We will address these, and more such pertinent questions in the next class.
- First we do some analysis in [R](#).

The Bordeaux equation and wine critics

- In 1991, Ashenfelter predicted 1989 and 1990 to be exceptional years. Wine critics did not agree at that time; but there is almost unanimous agreement now that they are the best vintages of the last 50 years.
- Ashenfelter's model and Parker's expert opinion both agree that the 2000 and 2003 vintage would be very high quality wine.
- Ashenfelter predicted the 1989 vintage to be excellent and 1990 even better. At first Robert Parker predicted this to be similar to the 1985 vintage but then later said it was the vintage of the century.
- All of this was done without tasting a single drop of wine.

How can it make my life better?

Top 10 Bordeaux Vintages

The top 10 Bordeaux vintages as predicted by the wine weather quality index is shown below.

Top 10 Bordeaux Vintages by Year

Based on a Wine Quality Weather Index

Year	Index Value
2003	1.128
2020	1.070
2018	1.061
2011	0.996
2006	0.856
2014	0.746
1961	0.721
2016	0.682
2019	0.602
1982	0.483

Top 10 Bordeaux Vintages based on the Wine Quality Index

The Ashenfelter wine quality index creates waves of controversy when it was first introduced. This post reevaluated the index while correcting for a few issues in an effort to improve it. Once reconstituted, we can calculate the top 10 Bordeaux vintages.

References

- Ashenfelter (1990), *Predicting the Quality and Prices of Bordeaux Wine*.
- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapter 3.
- James et al. (2023), *Introduction to Statistical Learning*, Chapter 3.